

Yitian Wu

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Tempe, US

EDUCATION

Shanghai Jiao Tong University (SJTU)

Shanghai, China

- M.S. in Mechanics, School of Aeronautics and Astronautics
- Relevant Courses: Finite Element Analysis of Composite, Solid Mechanics

09/22 – 09/25

GPA:3.46/4.0

Shanghai University (SHU)

Shanghai, China

- Bachelor of Mechatronics Engineering degree in Mechatronic Engineerin
- Relevant Courses: Theoretical Mechanics, Material Mechanics

09/18 – 07/22

GPA:3.42/4.0

RESEARCH EXPERIENCE

Establishment Test of Fatigue Strength Analysis for Wrinkled Fiber Structures in Aircraft Composite Materials

Research Assistant | Advisor: Yile Hu (SJTU)

01/23 – 08/25

- Fatigue tests are conducted on composite laminate structures containing folds of different sizes to obtain the corresponding mechanical properties. The fatigue fold permissible value and fiber fold structural overproof treatment method are established.
- Organizing and categorizing up to 729 specimens, designing and filling in over 243 sheets including Pre-test Measurement Record Sheets and NDT Record Sheets that related to test, processing test data and writing a 46000-word report on the phase of the test.
- First author paper 'Research On Fatigue Performance Analysis Method For Composite Material With Wrinkle Defect' is in progress.
- A utility model patent is pending: Improving the fixture of 'D7137/D7137M-23' standard based on the problems encountered during the test.

Experimental Investigation on the Joint Strength of Bonded Repairs in Aircraft Composite Structures

Research Assistant | Advisor: Yile Hu (SJTU)

10/22 – 08/23

- Determining the performance of adhesive joints under different adhesive repair methods, verifying the interface strength of adhesive repair structures, and validate repair analysis methods.
- Modelling 197 specimens and performing finite element analysis by using Abaqus to make a comparison to the test results.
- Recording pre-test dimensional measurements of 494 specimens, organizing various sheets related to test, processing test data and writing a 40000-word report on the completion of the project.

Experimental Investigation on Damage Characteristics of Representative Horizontal Stabilizer Structures

Research Assistant | Advisor: Yile Hu (SJTU)

10/23 – 01/25

- Assessment of Fatigue Damage Tolerance in Representative Structures on Aircraft Horizontal Stabilizer Surfaces.
- Establishing two types of Truss structure models via Abaqus software. Analyzing the models for linear buckling and post-buckling to help complete the load calibration of the fixtures.
- Designing a transverse static damage test fixture. Creating engineering drawings of long truss fixtures and shear fixtures of vertical/horizontal tails structures. Revising one atlas for each structure.

Flexural Evaluation of Repair Effectiveness in Composite Sandwich Structures

Research Assistant | Advisor: Yile Hu (SJTU)

10/22 – 01/23

- Comparing the bending stiffness and failure loads of defect-free and repaired specimens at room temperature and in hygrothermal environments. Determining the effectiveness of foreign repair materials and domestically produced repair materials.
- Participating in the writing of test reports and data processing.

PUBLICATION

[1] Y.T. Wu, W. Xi, Y.L. Hu. A Post-buckling Analysis Method for Transverse Deflection with Crest Shift in Curved Composite Laminates, Journal of Shanghai Jiao Tong University. (Accepted)

[2] Wu, Y., Hu, y., & Yu, y. "Post-impact compression fatigue fixture with slide structure. [Utility model patent]

SKILL

- Programming: Matlab, Python, Mathematica.
- Modeling: Solidworks.
- Finite element analysis: Abaqus.